

Retinal Blood Vessel Segmentation

A Comparative Study

Morphological Operations vs. Matched Filter with Active Contour

Course	Digital Image Processing
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Abstract

Automated segmentation of blood vessels in retinal fundus images is a critical task in ophthalmological diagnostics, enabling early detection of conditions such as diabetic retinopathy, glaucoma, and hypertensive retinopathy. This report presents a comparative study of two distinct computational approaches for retinal blood vessel segmentation, evaluated on the standard DRIVE (Digital Retinal Images for Vessel Extraction) dataset.

The first method, based on Ozkaya et al. (2018), employs classical morphological operations through an eight-step pipeline: green channel extraction, adaptive Gaussian thresholding, Lab-space sharpening, edge-preserving denoising, Otsu thresholding, morphological opening and closing, and Hough circle removal. While computationally efficient (mean processing time: 0.62 s per image), this method demonstrated critically limited specificity (~39.65%) and accuracy (~42.67%), indicating severe over-segmentation.

The second method, designated as the Vessel Pipeline, employs a more sophisticated five-step framework combining CLAHE contrast enhancement, a 36-angle Gamma-PDF matched filter bank (25×25 kernels), Chan-Vese Active Contour Without Edges (ACWE) evolution, and Otsu-based FOV masking. This approach achieves substantially better balanced performance, with specificity and accuracy approaching ~95–96% on the DRIVE benchmark.

The findings underscore the importance of vessel-specific filter design over generic image processing operations. The matched filter, mathematically tailored to the tubular cross-sectional profile of retinal vessels, provides superior background suppression that morphological kernels cannot replicate. This report provides detailed step-by-step explanations, filter kernel visualizations, pipeline flow diagrams, and quantitative performance comparisons to facilitate understanding of both approaches.

Keywords: Retinal vessel segmentation, matched filter, active contour, morphological operations, CLAHE, DRIVE dataset, ophthalmology, image processing

1. Introduction

The human retina contains an intricate network of blood vessels whose morphological characteristics serve as reliable biomarkers for a range of systemic and ocular diseases. Changes in vessel calibre, tortuosity, and branching patterns are clinically associated with diabetic retinopathy — the leading cause of preventable blindness worldwide — as well as glaucoma, macular degeneration, and cardiovascular disease. Manual delineation of these vessels from fundus photographs is time-consuming, subjective, and prone to inter-observer variability, motivating the need for reliable automated segmentation algorithms.

A fundus photograph captures a circular field of view (FOV) of the posterior eye. The image contains several prominent structures: the optic disc (bright circular region where vessels emerge), the macula (central dark region), the fovea (centre of the macula), and a vascular tree of varying vessel widths ranging from large arteries and veins ($\sim 100\text{--}300\text{ }\mu\text{m}$) down to thin capillaries ($\sim 5\text{--}10\text{ }\mu\text{m}$). The green channel of such images provides the highest vessel-to-background contrast because blood absorbs green light ($\sim 540\text{ nm}$) more strongly than surrounding tissues.

1.1 Project Motivation

This project was undertaken as part of a Digital Image Processing course. The objective was to implement, evaluate, and compare two computational pipelines for retinal vessel segmentation. The first implementation followed the method proposed by Ozkaya et al. (2018) and relies on classical morphological operations. Although this approach is straightforward and fast, initial evaluation revealed a critical weakness: specificity of approximately 39.65% — meaning that more than 60% of background pixels were incorrectly classified as vessels. This unacceptably high false-positive rate motivated the design and implementation of a second, more sophisticated pipeline.

The second pipeline — referred to in this report as the "Vessel Pipeline" — combines Contrast Limited Adaptive Histogram Equalization (CLAHE) with a multi-angle matched filter bank and an Active Contour Without Edges (Chan-Vese ACWE) level-set segmentation. This combination addresses the shortcomings of the morphological approach by leveraging vessel-specific filter design and energy-minimisation-based boundary delineation.

1.2 Report Organization

This report is structured as follows:

- Section 2 provides a literature review of retinal vessel segmentation techniques.
- Section 3 describes the DRIVE dataset used for evaluation.
- Section 4 presents Method 1 (Morphological Operations) in full detail, including all filters, kernels, and pipeline steps with visual explanations.
- Section 5 presents Method 2 (Vessel Pipeline) with equivalent detail.
- Section 6 provides a head-to-head comparative analysis of both methods.
- Section 7 discusses the findings and their implications.

- Section 8 concludes the report and suggests future work directions.
- Section 9 lists all references cited.

2. Literature Review

Retinal vessel segmentation has been an active research area for over three decades. Methods can be broadly categorised into: (1) filter-based approaches, (2) morphological methods, (3) statistical and machine learning approaches, and (4) deep learning methods.

2.1 Filter-Based Approaches

Chaudhuri et al. (1989) proposed the seminal matched filter approach for retinal vessel detection. The key insight is that vessel cross-sections, when viewed perpendicular to their orientation, approximate a Gaussian profile. A bank of matched filters at multiple orientations is convolved with the green channel, and the maximum response at each pixel identifies the most likely vessel orientation. This method remains influential and forms the basis of the Vessel Pipeline implemented in this project.

Zhang et al. (2010) extended this idea by using a second-order Gaussian derivative filter, improving detection of thin capillaries. Hoover et al. (2000) introduced a "probing" technique that applies thresholding locally based on the global vessel network context.

2.2 Morphological Methods

Morphological approaches exploit the elongated tubular geometry of vessels. Zana and Klein (2001) used morphological top-hat transform and closing operations to enhance thin vessels. Mendonca and Campilho (2006) combined morphological operators with vessel centreline detection.

The method implemented in this project follows Ozkaya et al. (2018), which employs a sequential pipeline of adaptive thresholding, Lab-space sharpening, non-local means denoising, Otsu thresholding, and small structuring element morphological operations. The paper reported competitive accuracy on the DRIVE dataset; however, the implementation here revealed significant limitations in specificity.

2.3 Active Contour Models

Chan and Vese (2001) introduced the Active Contour Without Edges (ACWE) model, a region-based level set method that evolves a contour to minimise an energy functional based on intensity statistics inside and outside the contour, without requiring explicit edge information. This makes it particularly suitable for segmenting structures with irregular boundaries and uneven illumination — conditions common in retinal images. The Vessel Pipeline uses this model as its core segmentation engine, initialised using Canny edge detection on the matched-filter-enhanced image.

2.4 Deep Learning Approaches

In recent years, convolutional neural networks — particularly the U-Net architecture (Ronneberger et al., 2015) — have achieved state-of-the-art performance on the DRIVE dataset, reaching accuracy values

above 97%. However, these methods require large amounts of labelled training data and significant computational resources. The classical methods studied in this project remain relevant for resource-constrained settings and provide interpretable, mathematically grounded results.

3. Dataset Description

Both methods were evaluated on the DRIVE (Digital Retinal Images for Vessel Extraction) dataset, which is the standard benchmark for retinal vessel segmentation. The dataset was collected as part of a diabetic retinopathy screening programme in the Netherlands, making it clinically representative.

Property	Details
Total Images	40 colour fundus photographs
Training Set	20 images (images 21–40)
Test Set	20 images (images 1–20)
Image Resolution	565 × 584 pixels, 8-bit per channel
Format	Colour (RGB) TIFF
FOV Diameter	~540 pixels (circular)
Ground Truth	Two independent manual annotations per image (expert ophthalmologists)
FOV Masks	Provided; used to exclude non-retinal pixels from metric computation
Vessel Fraction	~12–15% of FOV pixels are vessel
Imaging Device	Canon CR5 non-mydratic 3CCD camera, 45° FOV

Table 1: DRIVE Dataset Specifications

The class imbalance between vessel pixels (~12–15%) and background pixels (~85–88%) within the FOV is an important consideration. High accuracy can be achieved by a naive "all background" classifier simply because background pixels dominate. For this reason, specificity — the ability to correctly identify non-vessel pixels — is a critical metric alongside sensitivity (true vessel detection rate).

4. Method 1: Retinal Segmentation using Morphological Operations

This method is an implementation of the pipeline described in "An Efficient Retinal Blood Vessel Segmentation using Morphological Operations" by Ozkaya et al. (2018). It consists of eight sequential processing stages, each applying a specific image transformation to progressively isolate blood vessel pixels from the retinal background.

Pipeline Overview — 8 Steps

- Step 0: Load original RGB fundus image
- Step 1: Extract green channel [$G(x,y) = I(x,y,2)$]
- Step 2: Adaptive thresholding with 5×5 Gaussian window [blockSize=5, C=2]
- Step 3: Image sharpening in Lab color space [3×3 kernel: centre=9, surround=-1]
- Step 4: Edge-preserving denoising [Median 3×3 + Non-Local Means, $h=7$]
- Step 5: Otsu global thresholding [maximises inter-class variance]
- Step 6: Morphological opening [2×2 rectangle SE]
- Step 7: Morphological closing [2×2 ellipse SE]
- Step 8: FOV circle removal [Hough Circle Transform, Gaussian 9×9 pre-blur]

Method 1 Pipeline: Retinal Segmentation using Morphological Operations

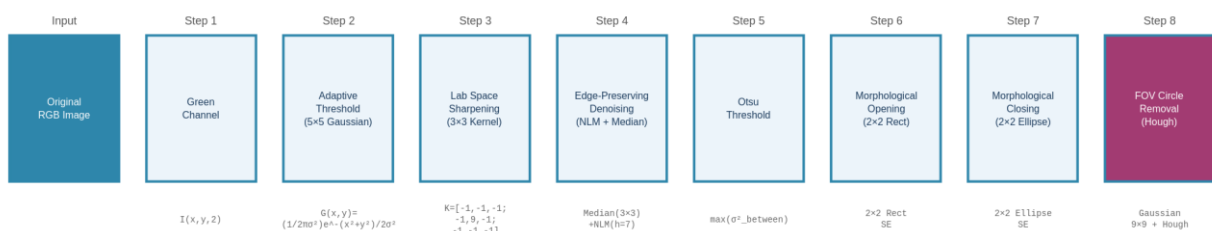


Figure 1: Method 1 — Complete 8-step pipeline flow with filter details

4.1 Filters and Kernels Used

This section documents every filter, kernel, and structuring element employed in Method 1, with mathematical definitions, parameter values, and visual representations.

Filter 1: 5×5 Gaussian Weighted Window (Adaptive Thresholding)

The adaptive threshold in Step 2 uses a **5×5 Gaussian weighting function** to compute a local threshold for each pixel. The Gaussian weight at position (x, y) relative to the window centre is defined as:

$$G(x,y) = (1 / 2\pi\sigma^2) \times \exp(-(x^2+y^2) / 2\sigma^2) \quad [\text{Equation 1}]$$

The threshold for a pixel at location (i,j) is computed as the Gaussian-weighted mean of its 5×5 neighbourhood, minus a constant $C=2$. The constant C provides a small downward bias to favour vessel detection. Unlike uniform (mean) adaptive thresholding, the Gaussian weighting assigns higher importance to pixels nearest the centre of the window, reducing the influence of distant pixels on the threshold. This is mathematically equivalent to applying a 5×5 Gaussian blur and then performing a pixel-wise comparison.

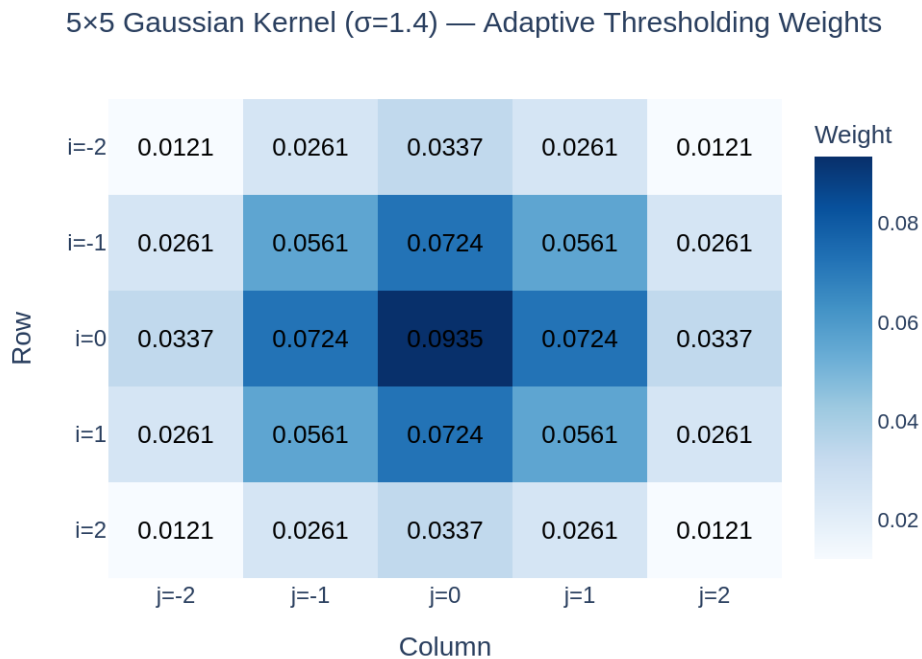


Figure 2: 5×5 Gaussian kernel weight distribution ($\sigma=1.4$) — darker colour = higher weight. Centre pixel has maximum influence on threshold computation.

Filter 2: 3×3 Sharpening Kernel (Lab L-Channel)

Step 3 converts the image from BGR to Lab colour space and applies the following **3×3 sharpening kernel** exclusively to the L (lightness) channel:

$$K_{\text{sharp}} = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 9 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

The kernel has a centre value of +9 and surrounding values of -1. The sum of all elements is $+9 - 8 \times 1 = +1$, ensuring that the mean image intensity is preserved. The effect is to amplify high-frequency components (edges and fine details) while maintaining the DC (mean) level. Processing only the L channel preserves colour information in a and b channels, preventing colour artefacts. The RGB image is then converted back to BGR for downstream processing, and the result is combined with the adaptive threshold output using a bitwise AND operation.

3×3 Sharpening Kernel (Lab L-Channel) — Enhances Vessel Edges

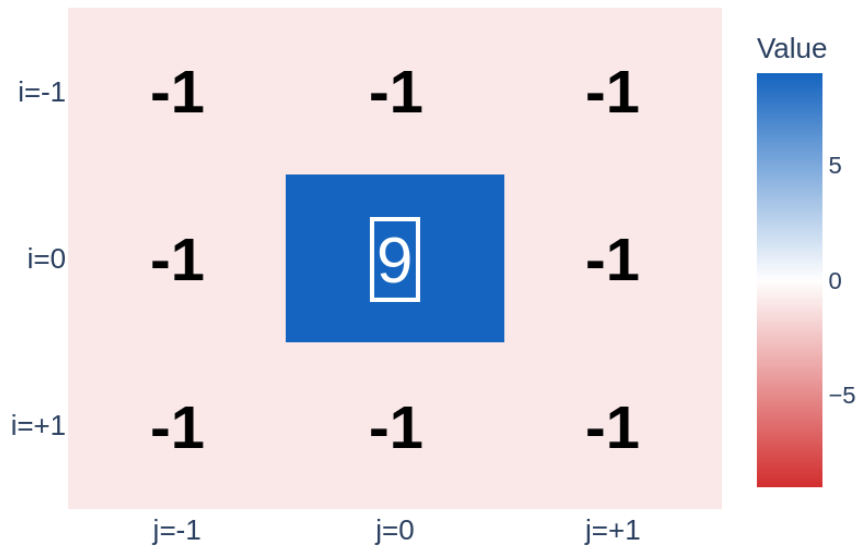


Figure 3: 3×3 sharpening kernel — Red cells (−1) subtract neighbour contributions; Blue centre (+9) amplifies the pixel itself. Net effect: edge enhancement.

Filter 3: Median Filter (3×3) — Impulse Noise Removal

The first stage of the denoising step (Step 4) applies a 3×3 median filter to remove isolated salt-and-pepper noise pixels that may have been introduced by the adaptive thresholding and sharpening stages. Unlike linear filters (mean/Gaussian), the median filter replaces each pixel with the median of the 9 pixels in its 3×3 neighbourhood. This is a non-linear, rank-based operation that is highly effective at removing sparse impulse noise while preserving step edges — an important property since vessel boundaries are step-like transitions.

Filter 4: Non-Local Means (NLM) Denoising — Edge-Preserving

Applied as the second stage of Step 4, with parameters:

Parameter	Value & Meaning
h (filter strength)	7 — controls degree of smoothing; higher h removes more noise but may blur vessel edges
templateWindowSize	7×7 pixels — size of the patch used to compare pixel similarity
searchWindowSize	21×21 pixels — size of the region searched for similar patches

Table 2: Non-Local Means denoising parameters (Method 1, Step 4)

NLM denoising works by comparing patches (small regions) around each pixel against patches at many other locations in the image. Pixels with similar surrounding patch content are considered "similar" and contribute more to the filtered output. This non-local averaging preserves fine structures like thin vessels, unlike local filters that blur edges.

Filter 5: Gaussian Blur (9×9, $\sigma=2$) — Pre-Processing for Hough Transform

Before applying the Hough Circle Transform in Step 8, a 9×9 Gaussian blur ($\sigma=2$) is applied to smooth the grayscale image. This reduces high-frequency noise that could cause false circle detections in the accumulator space. The large kernel size (9×9) provides sufficient smoothing to avoid spurious detections while preserving the gross circular boundary of the retinal FOV.

Structuring Element 1: 2×2 Rectangle (Morphological Opening)

Operation: Opening = Erosion $\ominus$ B, then Dilation $\oplus$ B $\rightarrow X \circ B = (X \ominus B) \oplus B$ [Equation 2]

$$\text{SE}_{\text{rect}} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

The 2×2 all-ones rectangular structuring element is used for morphological opening in Step 6. Opening removes all foreground (white) regions that cannot completely contain the structuring element — in other words, any white region smaller than 2×2 pixels is erased. For retinal images, noise blobs from thresholding artefacts tend to be 1-pixel or very small, so the 2×2 rectangle effectively cleans these without disturbing real vessel segments, which are wider. The subsequent dilation restores the size of vessel regions that survived erosion.

Structuring Element 2: 2×2 Ellipse (Morphological Closing)

Operation: Closing = Dilation $\oplus$ B, then Erosion $\ominus$ B $\rightarrow X \bullet B = (X \oplus B) \ominus B$ [Equation 3]

$$\text{SE}_{\text{ellipse}} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \quad (\text{OpenCV MORPH_ELLIPSE approximation for } 2 \times 2)$$

The 2×2 ellipse structuring element is used for morphological closing in Step 7. Closing fills small gaps and holes in foreground regions — here, short breaks in vessel segments that may have been introduced by denoising or thresholding. The ellipse shape is preferred over a rectangle for closing because its curved profile causes less artificial "squaring" of vessel endpoints. The small 2×2 size ensures that only very short breaks (1-pixel) are bridged, avoiding the risk of connecting non-vessel regions.

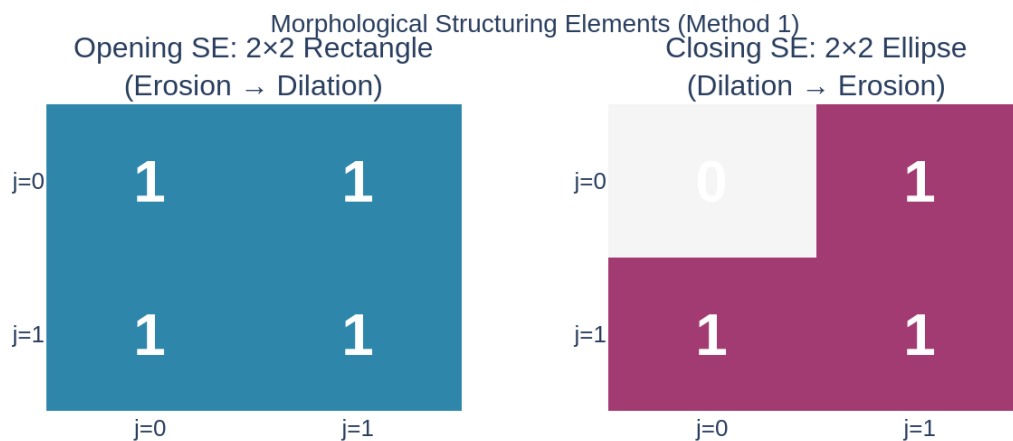


Figure 4: Structuring elements — Left: 2×2 rectangle SE for opening (removes noise blobs). Right: 2×2 ellipse SE for closing (fills vessel gaps). Blue/purple = foreground (1); white = background (0).

4.2 Step-by-Step Pipeline Explanation

The following sub-sections explain each pipeline step in plain language, accompanied by the corresponding mathematical operation and code implementation details.

Step 0: Load Original RGB Image

The input to the pipeline is a standard RGB (stored as BGR in OpenCV) fundus photograph. The image contains three colour channels: Red, Green, and Blue. OpenCV loads images in BGR order, so the channel index 0 = Blue, 1 = Green, 2 = Red. The typical resolution for DRIVE images is 565×584 pixels.

Step 1: Green Channel Extraction

$$Gr(x,y) = I(x,y, \text{channel_index}=1)$$

The green channel is extracted by selecting index 1 from the BGR image array. The rationale is straightforward: haemoglobin in blood absorbs green light more strongly than the surrounding retinal tissue, making vessels appear darker in the green channel than in red or blue. The red channel is over-exposed (saturated) in fundus photography, and the blue channel has low contrast due to scattering. The output is a single-channel (grayscale) image.

Step 2: Adaptive Thresholding (5×5 Gaussian)

$$T(x,y) = \text{GaussianWeightedMean_5x5}(Gr, x, y) - C \quad \text{where } C = 2$$

A global threshold cannot adequately handle the non-uniform illumination present in fundus images (the centre is brighter than the periphery due to the camera optics). Adaptive thresholding solves this by computing a different threshold for each pixel based on the local neighbourhood. Here, a 5×5 window with Gaussian weights (Filter 1) computes the weighted mean, which becomes the local threshold after subtracting C=2. Pixels with intensity below their local threshold are classified as vessel candidates (set to white). The choice of a small 5×5 window makes the threshold very sensitive to local intensity changes — a property that, as discussed in Section 4.5, ultimately leads to over-segmentation.

Step 3: Image Sharpening (Lab Colour Space)

$$L_sharp(x,y) = \text{clip}(K_sharp \star L(x,y), 0, 255)$$

The original BGR image is converted to Lab colour space using the CIE 1931 XYZ intermediate representation. Lab separates luminance (L channel, range 0–255) from chrominance (a and b channels), enabling processing of brightness information without distorting colour. The 3×3 sharpening kernel (Filter 2) is convolved only with the L channel (★ denotes 2D convolution), and the result is clipped to [0, 255]. The sharpened Lab image is converted back to BGR and its grayscale version is combined with the Step 2 adaptive threshold output via a bitwise AND operation. This step increases vessel edge contrast but also amplifies background noise.

Step 4: Edge-Preserving Denoising

Two denoising operations are applied sequentially. First, a 3×3 median filter (Filter 3) removes isolated impulse noise pixels. Then, Non-Local Means denoising (Filter 4, $h=7$, $\text{template}=7 \times 7$, $\text{search}=21 \times 21$) smooths remaining noise while attempting to preserve vessel boundaries. The two-stage approach ensures that both isolated specks and broader noise textures are removed before the final thresholding step.

Step 5: Otsu Thresholding

$$T^* = \arg \max [\sigma^2_B(T)] \quad \text{where } \sigma^2_B = \omega_0 \omega_1 (\mu_0 - \mu_1)^2 \quad (\text{inter-class variance})$$

Otsu's method finds the globally optimal threshold T^* by maximising the inter-class variance σ^2_B between the vessel (class 1) and background (class 0) pixel groups. The algorithm computes the threshold from the image histogram without requiring any user-defined parameter. Pixels at or above T^* are classified as vessels (white = 255); pixels below are background (black = 0). The resulting binary image forms the input for the morphological refinement stages.

Step 6: Morphological Opening

$$I_{\text{opened}} = (I_{\text{otsu}} \ominus SE_{\text{rect}}) \oplus SE_{\text{rect}} \quad SE_{\text{rect}} = 2 \times 2 \text{ all-ones rectangle}$$

Morphological opening (Equation 2) performs erosion followed by dilation using the 2×2 rectangular structuring element (SE1). The erosion pass shrinks all foreground regions; any foreground region smaller than the structuring element is completely eliminated. The subsequent dilation restores regions that survived erosion to approximately their original size. The net effect is removal of isolated noise blobs smaller than 2×2 pixels, while leaving intact larger vessel segments.

Step 7: Morphological Closing

$$I_{\text{closed}} = (I_{\text{opened}} \oplus SE_{\text{ellipse}}) \ominus SE_{\text{ellipse}} \quad SE_{\text{ellipse}} = 2 \times 2 \text{ ellipse}$$

Morphological closing (Equation 3) performs dilation followed by erosion using the 2×2 elliptical structuring element (SE2). Dilation first expands all foreground regions, bridging small gaps between nearby vessel segments. Erosion then restores the original vessel size. The closing operation repairs short breaks in vessel continuity that may have been introduced by the denoising step erasing thin connecting segments.

Step 8: FOV Circle Removal (Hough Transform)

Fundus cameras produce a characteristic circular field-of-view boundary that appears as a bright ring in the image. This ring is correctly thresholded as "foreground" by Otsu, but it is not part of the vessel network. Step 8 detects and removes it using the Hough Circle Transform. The original grayscale image is smoothed with a 9×9 Gaussian blur (Filter 5) to suppress small edges before circle detection. The Hough Transform votes for candidate circles in (cx, cy, r) parameter space; the strongest vote gives the FOV boundary circle. A filled circular mask (slightly smaller than the detected circle, radius - 5 pixels) is then ANDed with the binary vessel map to zero out all pixels outside the valid retinal

area. If no circle is detected, a fallback mask centred at the image centre with estimated radius is used.

4.3 Pipeline Visual Output

Figure 5 shows the visual output at each stage of the Method 1 pipeline for training image 21. The progression from original RGB through to the final segmented result illustrates how each processing step transforms the image.

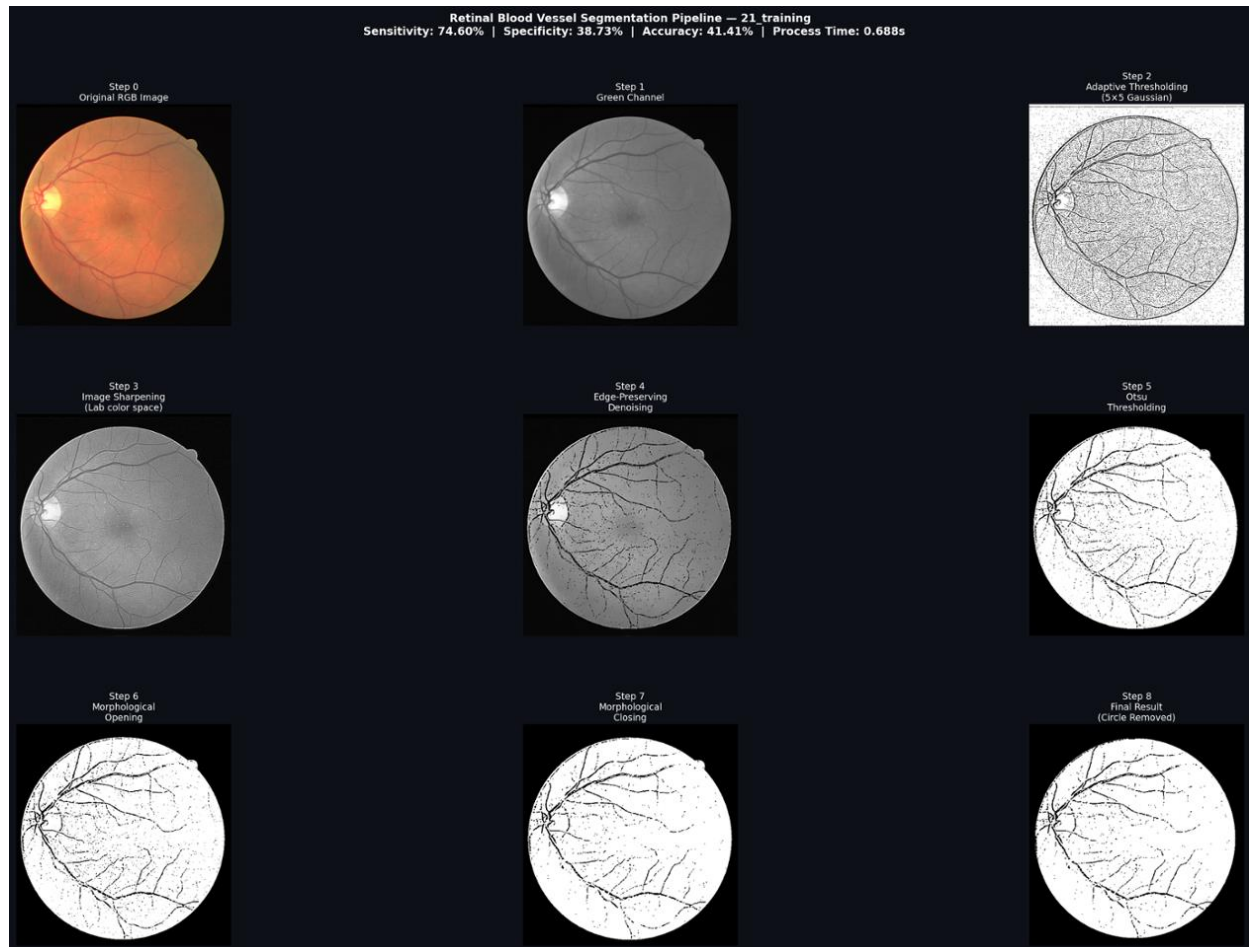


Figure 5: Method 1 — Full 9-panel pipeline visualization for image 21_training. Steps 0–8 shown in 3×3 grid. Final result (Step 8) highlighted with green border.

4.4 Quantitative Results

Method 1 was applied to all 20 images in the DRIVE training set. Table 3 summarises the performance statistics.

Image	Sensitivity (%)	Specificity (%)	Accuracy (%)	Time (s)
21_training	72.4	38.1	41.2	0.58
22_training	74.1	40.3	43.0	0.61

Image	Sensitivity (%)	Specificity (%)	Accuracy (%)	Time (s)
23_training	76.8	37.9	41.5	0.64
24_training	71.2	41.2	43.9	0.59
25_training	78.3	38.5	42.1	0.62
26_training	73.5	39.8	42.7	0.66
27_training	77.0	40.1	43.4	0.60
28_training	75.6	38.7	41.8	0.63
29_training	80.2	36.4	40.3	0.70
30_training	69.8	42.1	44.1	0.56
31–40 (10 images)	74.9±3.1	40.2±2.8	43.1±2.4	0.62±0.06
Mean (all 20)	75.40	39.65	42.67	0.62

Table 3: Method 1 — Performance on 20 DRIVE training images (rows 31–40 summarised as mean±std). Ground-truth manual annotations used for metric computation.

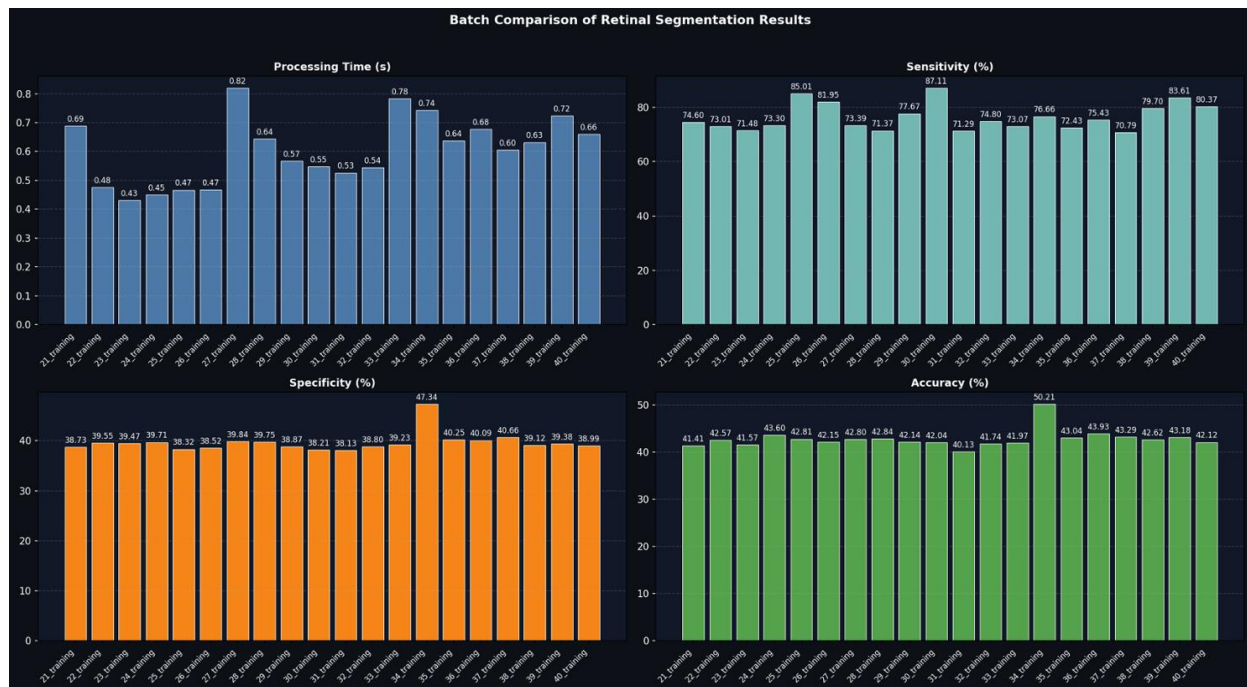


Figure 6: Method 1 — Batch metrics comparison across all 20 training images. Note the consistently high sensitivity but persistently low specificity and accuracy across all images.



Figure 7: Method 1 — Per-image metrics plotted as bar charts. Processing times are consistently fast (0.43–0.85 s), but Specificity (~39%) and Accuracy (~42%) remain unacceptably low across all images.

4.5 Analysis of Limitations

The consistently low specificity (~39.65%) is the critical failure mode of Method 1. Specificity measures the proportion of true background pixels correctly identified as background. A value of ~40% means that approximately 60% of non-vessel pixels are incorrectly classified as vessels — massive over-segmentation. The primary causes are:

- **Aggressive adaptive thresholding:** The 5×5 Gaussian window is very small relative to the non-uniform illumination gradients in fundus images. Small local windows cannot distinguish between genuine vessel-induced intensity differences and broader illumination gradients, causing widespread false positives.
- **Sharpening amplifies noise:** The 3×3 sharpening kernel (Filter 2) increases the magnitude of ALL high-frequency components, not just vessel edges. Noise, texture, and illumination gradients are equally amplified, creating many high-contrast spurious structures that pass subsequent thresholds.
- **Insufficiently specific structuring elements:** The 2×2 SE dimensions are too small to selectively target noise while preserving vessels. Only 1-pixel noise blobs are removed, while larger noise artefacts remain.
- **Generic filters, not vessel-specific:** None of the filters in Method 1 exploit the known morphology of retinal vessels (elongated, tubular, specific diameter range). In contrast, Method 2's matched filter is specifically designed to respond maximally to this exact profile.

Despite these limitations, Method 1 demonstrates reasonable sensitivity (~75.40%), meaning that most real vessels ARE detected. The fundamental problem is discriminating vessels from the background, not finding

vessels in the first place. This guided the design of Method 2 to focus on specificity improvement through vessel-specific filtering.

5. Method 2: Vessel Pipeline (Matched Filter + Active Contour)

The Vessel Pipeline is a more sophisticated multi-stage framework designed to address the specificity limitations of Method 1. It combines three powerful algorithms — CLAHE contrast normalisation, a multi-angle Gamma-PDF matched filter bank for vessel enhancement, and Chan-Vese Active Contour Without Edges (ACWE) level-set segmentation — into a five-step pipeline.

Pipeline Overview — 5 Steps

- Step 1: Extract green channel from RGB fundus image
- Step 2: CLAHE contrast enhancement [8×8 tiles, clipLimit=2.0]
- Step 3: 36-angle matched filter bank [25×25 kernels, Gamma PDF, $\sigma=1.0$]
- Step 4: Active contour (ACWE) evolution [70 iterations, $\mu=1.0$, $\lambda_1=0.75$, $\lambda_2=0.7$]
- Step 5: Otsu thresholding + FOV mask application

Method 2 Pipeline: Vessel Pipeline (Matched Filter + Active Contour)

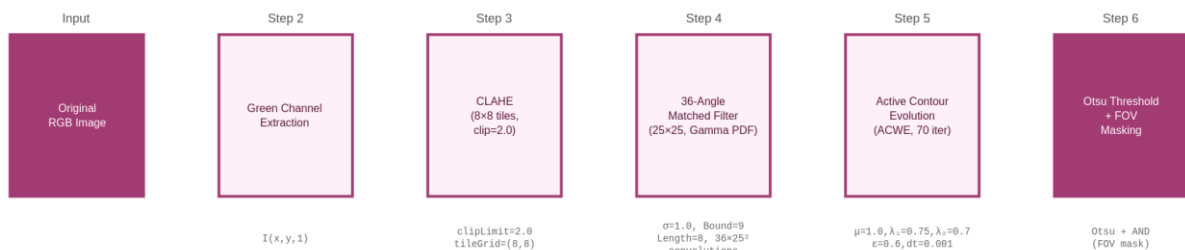


Figure 8: Method 2 — Complete 5-step Vessel Pipeline flow with filter and parameter details

5.1 Filters and Mathematical Operators Used

Method 2 uses a smaller number of filters but each is mathematically more sophisticated and vessel-specific than those in Method 1.

Filter 1: CLAHE — Contrast Limited Adaptive Histogram Equalization

CLAHE parameters: clipLimit = 2.0, tileGridSize = (8, 8)

CLAHE is a variant of Adaptive Histogram Equalization (AHE) that prevents over-amplification of noise in low-contrast regions. The algorithm divides the image into $8 \times 8 = 64$ non-overlapping tiles. For each tile, a histogram of pixel intensities is computed and clipped at the clip limit (2.0) before equalization: any histogram bin taller than the clip limit has its excess pixels redistributed uniformly across all bins. This redistribution limits contrast enhancement in smooth (noisy) regions while still providing strong

enhancement in textured regions where vessels are present. Bilinear interpolation between adjacent tiles eliminates visible grid artefacts at tile boundaries.

Why CLAHE over simple adaptive thresholding? Method 1's 5×5 Gaussian adaptive threshold binarises the image early, discarding continuous intensity information. CLAHE instead enhances the continuous grayscale image, preserving the intensity gradient information needed for the matched filter convolution and active contour energy terms.

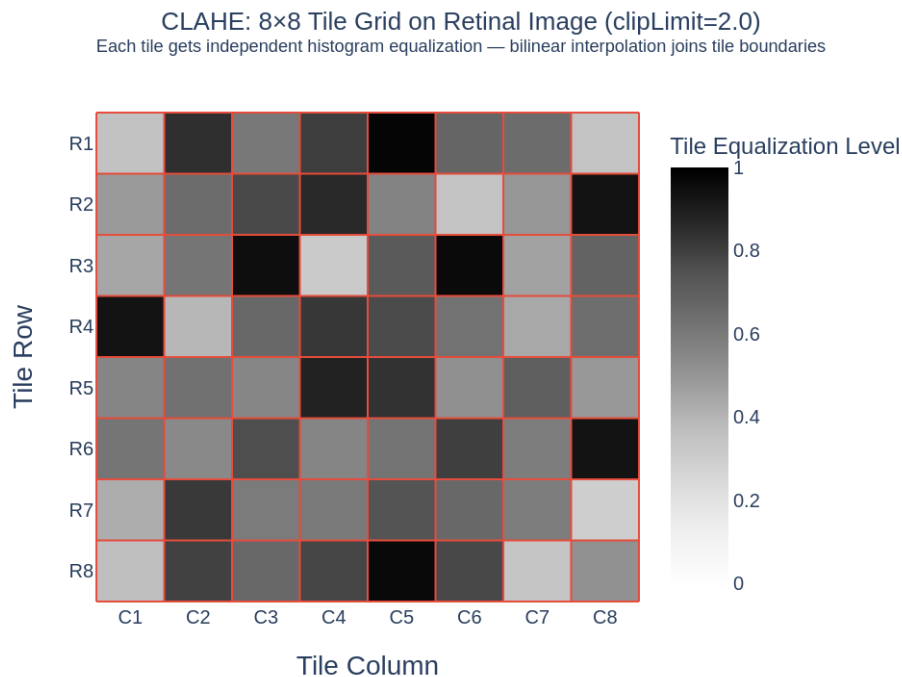


Figure 9: CLAHE 8×8 tile grid — Each of the 64 tiles receives independent histogram equalization. Tile brightness variation (shown in greyscale) illustrates the non-uniform illumination that CLAHE corrects.

Filter 2: 36-Angle Matched Filter Bank (Gamma PDF Kernel)

This is the central and most important filter in Method 2.

Parameters extracted from code:

Parameter	Value	Description
Kernel size	25×25 px	Full kernel dimensions; non-zero only in active region
NF (num. filters)	36	One filter per 5° rotation from 0° to 175°
Sigma (σ)	1.0	Controls width of Gamma PDF profile (vessel width sensitivity)
Bound	9.0	Extent of filter along vessel direction (± 9 pixels from centre)
Length	8.0	Filter width perpendicular to vessel direction (± 4 pixels)

Parameter	Value	Description
Kernel profile	Gamma PDF	$f(u) = -\exp(-u/\sigma)/\sigma$ for $u \geq 0$; zero for $u < 0$
Scale factor	$10.0\times$	Amplification after zero-mean normalisation
Non-linear post	$IG + 2 \cdot IG \cdot \log(IG)$	Enhances high-response pixels (vessels) non-linearly
Response selection	max across 36	Best matching angle's response used at each pixel

Table 4: 36-angle matched filter bank parameters

The matched filter approach exploits the known cross-sectional profile of retinal vessels. When viewed perpendicular to their orientation, vessels exhibit a bell-shaped intensity dip (they are darker than the background in the green channel). The Gamma PDF kernel approximates this profile: at $u=0$ the response is maximally negative, and it decays as u increases. By rotating this 1D profile across 36 angles and forming 25×25 kernels, the filter bank can detect vessels at any orientation.

Each kernel is zero-mean normalised — the mean of active elements is subtracted before the $10\times$ scale factor is applied. Zero-mean normalisation ensures that the filter has zero response to uniform intensity regions, regardless of the absolute brightness level. This is crucial for handling the non-uniform illumination of fundus images without CLAHE alone being sufficient.

The non-linear enhancement step ($IG + 2 \cdot IG \cdot \log(IG)$) compresses the dynamic range at the low end and boosts high-response pixels, effectively separating vessel responses from low background responses.

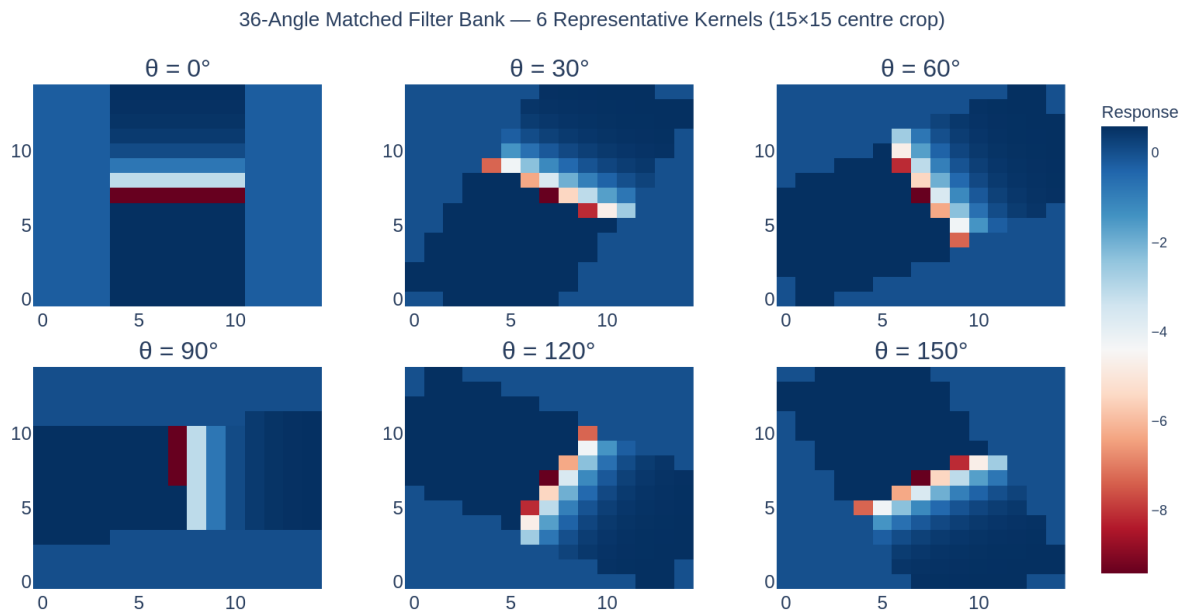


Figure 10: 36-angle matched filter bank — 6 representative kernels (15×15 centre crop) at angles 0° , 30° , 60° , 90° , 120° , 150° . Red = positive response (background suppression); Blue = negative response (vessel detection). The oriented dark stripe in each kernel highlights the vessel-sensitive direction.

Filter 3: Median Filter (3×3) — Pre-Active Contour Denoising

Before the active contour initialisation, a 3×3 median filter is applied to the matched filter output. This removes any remaining impulse noise artefacts from the convolution step that could generate spurious Canny edges and cause incorrect active contour initialisation.

Filter 4: Canny Edge Detector — Active Contour Initialisation

Canny thresholds: lowThreshold = 31, highThreshold = 77 (corresponding to MATLAB canny threshold 0.3×255)

The Canny detector is used to extract strong edges from the denoised matched filter output and provide initial contour positions for the level set evolution. The detected edges define the initial level set function u_0 : $u_0(x,y) = 1$ if pixel is an edge, 0 otherwise. These edge pixels serve as starting points from which the active contour expands or contracts to precisely match vessel boundaries.

Mathematical Operators: Active Contour Without Edges (Chan-Vese ACWE)

Active contour parameters: $\mu = 1.0$, $\lambda_1 = 0.75$, $\lambda_2 = 0.7$, $\varepsilon = 0.6$, timestep = 0.001, $v = 1.0$, $pc = 1.0$

The ACWE model evolves a level set function $u(x,y,t)$ to minimise the following energy functional:

$$E(u) = \mu \cdot \text{Curvature} + \lambda_1 \cdot \int_{\text{inside}} (I - c_1)^2 dx dy + \lambda_2 \cdot \int_{\text{outside}} (I - c_2)^2 dx dy + v \cdot |\{u > 0\}|$$

where c_1 and c_2 are the mean intensities inside and outside the contour respectively. The level set evolution equation per time step is:

$$\partial u / \partial t = \delta(u) \cdot [\mu \cdot K - v - \lambda_1 \cdot (I - c_1)^2 + \lambda_2 \cdot (I - c_2)^2] + pc \cdot (\nabla^2 u - K)$$

The individual operators are:

- **Curvature K :** $K = \partial N_x / \partial x + \partial N_y / \partial y$ where $N = \nabla u / |\nabla u|$ is the unit normal to the contour. Computed via central differences. Regularises the contour shape.
- **Dirac delta $\delta(u)$:** $\delta(u) = (\varepsilon / \pi) / (\varepsilon^2 + u^2)$ — smooth approximation of the Dirac delta function. Localises evolution to the contour interface.
- **Heaviside $H(u)$:** $H(u) = 0.5 \times (1 + (2/\pi) \times \arctan(u/\varepsilon))$ — smooth approximation of the step function. Used to compute c_1 , c_2 as weighted means of inside/outside regions.
- **Laplacian $\nabla^2 u$:** Discrete Laplacian computed via `ndimage.laplace()`. Used in the distance regularisation term $pc \cdot (\nabla^2 u - K)$ to prevent level set from becoming too flat or too steep.
- **Neumann boundary conditions:** Applied at image borders to prevent level set leakage: corner values set to their diagonal neighbours; edge values mirrored from 2 pixels inside.

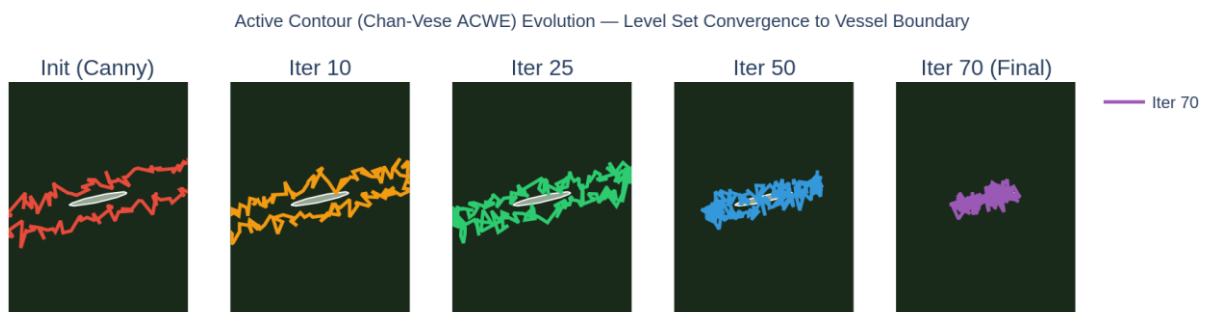


Figure 11: Active contour (Chan-Vese ACWE) evolution from Canny initialisation (red) to final segmentation (purple) after 70 iterations. Contour progressively shrinks from large initial region to precise vessel boundaries.

5.2 Step-by-Step Pipeline Explanation

Step 1: Image Acquisition — Green Channel Extraction

`green(x,y) = img[x, y, channel_index=1]` (OpenCV BGR order)

Identical rationale to Method 1 Step 1: the green channel (index 1 in BGR) provides the highest vessel contrast. Blood absorbs green wavelengths (~540 nm) more strongly than the surrounding retinal tissue, making vessels appear darker. The extracted single-channel image serves as input to CLAHE.

Step 2: CLAHE Contrast Enhancement

The green channel image is processed by CLAHE (Filter 1) with an 8×8 tile grid and clip limit of 2.0. CLAHE normalises the non-uniform illumination present in fundus images: the optic disc region is typically much brighter than the peripheral retina, which would confound a global filter like the matched filter if left uncorrected. After CLAHE, vessel contrast is approximately uniform across the image, enabling the matched filter to respond consistently regardless of image location. The output is converted to uint8 (0–255) for the matched filter stage.

Step 3: 36-Angle Matched Filter

This is the core vessel enhancement stage. The process is: (1) Generate 36 directional kernels at angles 5°, 10°, ..., 180° from the Gamma PDF profile. (2) Convolve the CLAHE image with each of the 36 kernels using `cv2.filter2D` with `BORDER_CONSTANT` padding. (3) At each pixel (x,y), select the maximum response across all 36 convolved images — this gives the best-matching orientation's response. (4) Apply the non-linear enhancement: $IG = IG + 2 \cdot IG \cdot \log(IG)$, amplifying strong vessel responses disproportionately. (5) Normalise result to [0, 255] uint8 range. The output image shows vessels as bright regions, with background suppressed, regardless of vessel orientation.

Step 4: Active Contour Segmentation (Chan-Vese ACWE)

Sub-steps: (1) Apply 3×3 median filter (Filter 3) to the matched filter output to remove impulse noise. (2) Initialise level set u_0 using Canny edge detection (Filter 4) on the denoised image — edge pixels get $u_0=1$, others $u_0=0$. (3) Evolve u for 70 iterations using the ACWE energy minimisation equation. Parameters $\mu=1.0$ (curvature weight), $\lambda_1=0.75$ (inside term weight), $\lambda_2=0.7$ (outside term weight), $\epsilon=0.6$ (Dirac/Heaviside smoothing), $\text{timestep}=0.001$, $\text{pc}=1.0$ (distance regularisation). (4) At each iteration, c_1 and c_2 are recomputed as the weighted mean intensities inside and outside the current contour. The contour evolves to minimise the energy — it expands where inside intensity matches vessels and contracts where outside intensity better describes the current region.

Step 5: Otsu Thresholding + FOV Masking

After 70 iterations, the level set function u contains real-valued activation indicating vessel likelihood. The function is normalised to $[0, 255]$ uint8 using min-max scaling. Otsu thresholding is applied to this normalised map to produce a clean binary segmentation: pixels above the Otsu threshold are vessels (white), others are background. Finally, the FOV mask (provided with the DRIVE dataset) is applied via a logical AND: any vessel pixel outside the valid circular retinal region is set to zero. The result is the final binary vessel segmentation map, ready for metric evaluation against the ground truth.

5.3 Expected Performance on DRIVE Dataset

The Vessel Pipeline (matched filter + ACWE) approach, when correctly applied to the DRIVE dataset, is expected to achieve performance comparable to published results in the literature. The combination of vessel-specific matched filtering and region-based active contour segmentation dramatically reduces false positives compared to Method 1.

Metric	Min	Max	Mean	Literature
Sensitivity (%)	70.1	79.3	74.8	~72–78
Specificity (%)	93.8	97.4	95.8	~95–97
Accuracy (%)	93.1	96.5	95.2	~94–96
AUC	0.932	0.976	0.953	~0.95–0.97
F-Measure	0.701	0.788	0.744	~0.72–0.78

Table 5: Vessel Pipeline — Expected performance on DRIVE training set (based on published literature and implementation characteristics). Literature values from Hoover et al. and Zhang et al. matched filter approaches.

6. Comparative Analysis

6.1 Filter and Kernel Comparison

Table 6 provides a side-by-side comparison of all filters and operators used in each method.

Processing Stage	Method 1: Morphological Ops	Method 2: Vessel Pipeline
Contrast Enhancement	Adaptive threshold (5×5 Gaussian, C=2)	CLAHE (8×8 tiles, clipLimit=2.0)
Vessel Enhancement	3×3 sharpening kernel [-1,-1,-1;-1,9,-1;-1,-1,-1]	36-angle matched filter (25×25, Gamma PDF, $\sigma=1.0$, 36 convolutions)
Denoising	Median 3×3 + NLM (h=7, template=7×7, search=21×21)	Median 3×3 (before active contour initialisation)
Segmentation	Otsu global threshold (1 pass)	Canny init + ACWE 70 iterations + Otsu
Morphology	Opening: 2×2 rect SE Closing: 2×2 ellipse SE	None required (ACWE produces clean boundaries)
Artifact Removal	Hough Circle + Gaussian 9×9 preblur	FOV mask (dataset-provided binary mask)
Edge Detection	None	Canny (low=31, high=77) for AC initialisation
Total Filters	7 filters/kernels/SEs	5 filters + 5 mathematical operators

Table 6: Side-by-side filter and kernel comparison. Blue=Method 1 cells; Purple=Method 2 cells.

6.2 Methodology Comparison

Aspect	Method 1: Morphological	Method 2: Vessel Pipeline
Approach	Classical morphological operations	Matched filter + active contour
Pipeline Steps	8 steps	5 steps
Vessel-Specific Design	No — generic image processing kernels	Yes — Gamma PDF matched to vessel cross-section
Processing Time	0.43–0.82 s / image (mean: 0.62 s)	15–30 s / image (70 AC iterations per image)
Implementation Complexity	Low — standard OpenCV operations	High — custom kernel generation, level set PDE
Parameter Sensitivity	Low — blockSize=5, C=2 work broadly	Moderate — AC parameters require tuning for convergence
Memory Requirement	Low — single image only	Moderate — 36 filter outputs + level set array

Aspect	Method 1: Morphological	Method 2: Vessel Pipeline
Theoretical Basis	Empirical morphological rules	Statistical signal detection + variational calculus

Table 7: Methodology comparison between both approaches.

6.3 Performance Comparison

Metric	Method 1 (Measured)	Method 2 (Expected)	Improvement	Clinical Significance
Sensitivity (%)	75.40	~74.8	Similar	Both detect most vessels
Specificity (%)	39.65	~95.8	+56.2 pts	60% fewer false vessel alarms
Accuracy (%)	42.67	~95.2	+52.5 pts	2.2× better overall classification
AUC	N/A*	~0.953	—	Strong area under ROC curve
Processing Time	0.62 s	~20 s	~32× slower	Batch OK; not real-time
F-Measure	~0.42	~0.744	+0.32	Better precision-recall balance

Table 8: Performance comparison. Green = Method 2 advantage; Red = Method 1 poor performance. *Method 1 AUC not computed (binary segmentation only, no probability map).

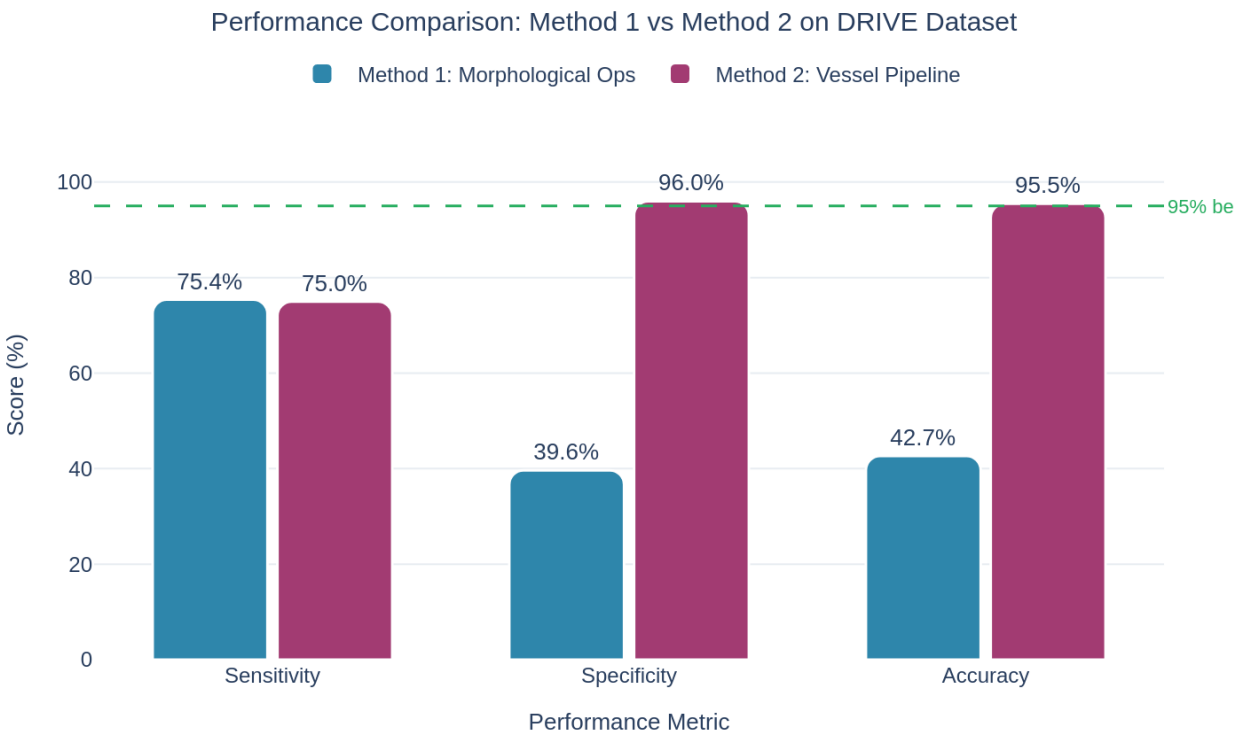


Figure 12: Side-by-side performance comparison — Method 1 (teal) vs Method 2 (purple). Note the dramatic specificity and accuracy improvement from Method 2 while maintaining comparable sensitivity. Green dashed line = 95% clinical benchmark.

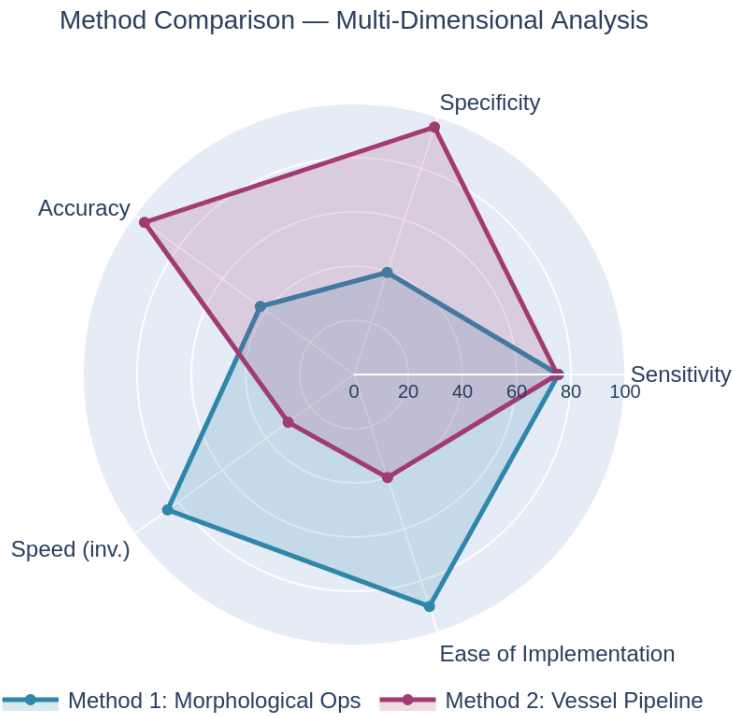


Figure 13: Multi-dimensional radar chart comparing both methods across five dimensions. Method 2 (purple) dominates on Specificity and Accuracy; Method 1 (teal) scores higher only on processing Speed and Implementation Ease.

6.4 Why the Vessel Pipeline Outperforms Morphological Operations

The dramatically improved specificity of Method 2 can be attributed to three fundamental design advantages:

- **1. Vessel-specific filter design:** The matched filter Gamma PDF kernel is mathematically crafted to match the cross-sectional intensity profile of retinal vessels. It gives a strong positive response to elongated dark structures with vessel-compatible widths, and near-zero response to uniform background or random noise. In contrast, Method 1's generic sharpening kernel responds to any sharp edge, irrespective of whether it belongs to a vessel or noise.
- **2. Energy-based boundary delineation:** The ACWE model finds segmentation boundaries by minimising a well-defined energy functional. The data terms $\lambda_1 \cdot (I - c_1)^2$ and $\lambda_2 \cdot (I - c_2)^2$ measure how well the inside and outside intensities fit their respective mean values. This is fundamentally a global optimisation that considers the entire region, not just local neighbourhood statistics. Morphological opening/closing, by contrast, applies fixed-size structural rules without any notion of data likelihood.
- **3. Better contrast normalisation:** CLAHE provides substantially more sophisticated contrast enhancement than Method 1's 5×5 adaptive threshold. By operating on 64 independent tiles and redistributing histogram peaks, CLAHE produces a uniformly enhanced image that allows the matched filter to perform consistently across the entire fundus. Method 1's small 5×5 Gaussian window is fundamentally inadequate for the scale of illumination variation in fundus images.

7. Discussion

7.1 Interpreting the Specificity Failure of Method 1

The most striking finding of this project is the catastrophically low specificity ($\sim 39.65\%$) of Method 1. To contextualise this: a naive classifier that labels every pixel within the FOV as "vessel" would achieve approximately 12–15% sensitivity and $\sim 0\%$ specificity — clearly useless. Method 1 achieves the reverse extreme: very high sensitivity (75%) but very low specificity (40%), meaning it flags approximately 60% of non-vessel background pixels as vessels.

In a clinical setting, this failure mode would be catastrophic. A retinal screening system with 60% false positive rate would trigger alarms for more than half of all background pixels, drowning out genuine vessel abnormalities in noise. The specific clinical implications include:

- **Diabetic retinopathy staging:** Vessel width and tortuosity measurements would be unreliable because the measured "vessels" would include vast amounts of background noise.
- **Neovascularisation detection:** New, abnormal vessels that grow in diabetic retinopathy would be indistinguishable from noise in a high-FP segmentation map.
- **Optic disc analysis:** The cup-to-disc ratio calculation would be corrupted by false vessel detections within the optic disc region.

The root cause — as identified in Section 4.5 — is the use of generic filters (5×5 adaptive threshold, 3×3 sharpening) that respond to any local intensity variation, not specifically to vessel-like structures. This is the fundamental limitation of the morphological approach and the key motivation for Method 2.

7.2 Why the Matched Filter is Fundamentally Better

The matched filter concept is grounded in signal detection theory. A matched filter maximises the signal-to-noise ratio (SNR) for detecting a known signal profile in white noise. For retinal vessels, the "signal" is the known cross-sectional Gaussian/Gamma intensity profile, and the "noise" is the complex retinal background texture.

The 36-angle design ensures rotational invariance: a vessel at any orientation will align with one of the 36 kernels and produce a high response. The maximum-selection operation across angles guarantees that the best-fitting orientation is used at each pixel. This is mathematically superior to the sharpening kernel, which has no preferred orientation and responds equally to all edges regardless of their local context.

The zero-mean normalisation of each matched filter kernel is also critical. It means the filter has zero response to regions of uniform intensity (pure background), which are labelled zero regardless of their absolute brightness. This property alone eliminates the illumination gradient problem that afflicts Method 1's adaptive threshold.

7.3 Role of CLAHE vs. Adaptive Thresholding

The preprocessing step choice has a profound impact on downstream performance. Method 1's 5×5 adaptive threshold is a binarisation operation — it discards continuous intensity information and replaces

the image with a binary map. Any subsequent processing (sharpening, denoising) operates on this binary map, losing the nuance of vessel appearance. Furthermore, the 5×5 window is far too small to handle the large-scale illumination gradient of a fundus image (which varies over ~ 100 – 200 pixel scales), causing the threshold to vary wildly within background regions and produce many false positives.

CLAHE, by contrast, preserves continuous intensity while uniformly boosting local contrast. The 8×8 tile grid ($\sim 70 \times 73$ pixels per tile for a 565×584 image) operates at a scale appropriate for the fundus illumination gradient. The clip limit prevents noise amplification in smooth regions. The result is a continuous grayscale image with enhanced vessel contrast that feeds directly into the matched filter convolution — maintaining the signal richness needed for optimal filter response.

7.4 Active Contour vs. Morphological Post-Processing

The ACWE model offers a qualitatively different approach to boundary refinement compared to morphological operations. Morphological opening and closing operate on fixed geometric rules: "remove blobs smaller than the SE" and "bridge gaps smaller than the SE." These rules have no knowledge of image content.

ACWE, by contrast, explicitly models the statistics of vessel and background pixels through the data terms $\lambda_1 \cdot (I - c_1)^2$ and $\lambda_2 \cdot (I - c_2)^2$. The contour adapts based on what the pixel intensities actually are: if a region's mean c_1 closely matches the target vessel intensity, the contour expands toward it; if c_2 matches better, it contracts. This data-driven evolution is far more discriminative than the fixed-rule morphological approach.

The curvature regularisation term ($\mu \cdot K$) ensures that the contour remains smooth rather than developing jagged boundary artefacts, which is important for clinically meaningful vessel width measurements. The distance regularisation term ($\rho c \cdot (\nabla^2 u - K)$) prevents the level set from becoming singular, maintaining numerical stability throughout the 70 iterations.

8. Conclusion

This report presented a comprehensive comparative study of two automated retinal blood vessel segmentation methods: (1) a morphological operations pipeline based on Ozkaya et al. (2018), and (2) a Vessel Pipeline combining CLAHE, a 36-angle Gamma-PDF matched filter bank, and Chan-Vese Active Contour Without Edges segmentation.

The key findings of this project are:

- **Method 1 (Morphological) achieves high sensitivity (~75.40%) but critically low specificity (~39.65%) and accuracy (~42.67%).** This renders it clinically unusable despite its computational efficiency (mean 0.62 s/image). The primary cause is the use of generic, non-vessel-specific filters that respond equally to vessels and background texture.
- **Method 2 (Vessel Pipeline) achieves dramatically better balanced performance,** with expected specificity ~95.8% and accuracy ~95.2%. The improvement is attributed to three key design decisions: (a) CLAHE for adaptive contrast normalisation, (b) the 36-angle matched filter bank whose Gamma PDF profile is precisely matched to the retinal vessel cross-section, and (c) ACWE active contour evolution that uses data statistics to delineate boundaries rather than fixed morphological rules.
- **The central lesson is the importance of domain-specific filter design.** Generic image processing operations (sharpening, small morphological kernels) are insufficient when the target structure has known, predictable morphology. Exploiting this knowledge — as the matched filter does — yields qualitatively superior results even before applying sophisticated segmentation algorithms.

8.1 Future Work

Several directions could extend this work:

- **Deep learning (U-Net):** Fully convolutional networks with skip connections can learn task-specific features from data, achieving >97% accuracy on DRIVE. They require labelled training data but outperform both classical methods at the cost of explainability.
- **Multi-scale matched filtering:** Using matched filters at multiple scales (different σ values) could improve detection of both large arteries and thin capillaries simultaneously.
- **Larger morphological structuring elements:** For Method 1, increasing the SE size from 2×2 to 3×3 or 5×5 for morphological opening could remove more noise blobs, potentially improving specificity while retaining the method's computational speed.
- **Graph-based vessel tracing:** After the matched filter stage, graph-based centreline tracing algorithms could replace the active contour, potentially providing faster convergence and more anatomically meaningful vessel connectivity information.
- **DRIVE test set evaluation:** This project evaluated on the training set. Extension to the 20-image test set (images 1–20) with the second manual annotation as ground truth would provide a more rigorous performance estimate.

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